

WASP-140b

R-2361

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_200109 · created 2026-08-19T23:30:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458857.6575 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.05
Transit depth (Rp/Rs)^2	1.65 +/- 1.27 [%]
Orbital Inclination (inc)	83.01 +/- 2.89
Airmass coefficient 1 (a1)	0.853 +/- 0.015
Airmass coefficient 2 (a2)	0.1 +/- 0.3
Scatter in the residuals of the lightcurve fit is	1.74 %
Best Comparison Star	#1 - [617, 348]
Optimal Method	PSF photometry
Transit Duration (day)	0.045 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.129 ± 0.05 vs 0.1432 (deviation 0.28σ)
Duration fit vs published	0.045 d vs 0.0604 d (deviation 0.41σ)
Mid-time O–C	-1.2 min
Depth SNR	2.6
Residual scatter	1.74 %

Light curve

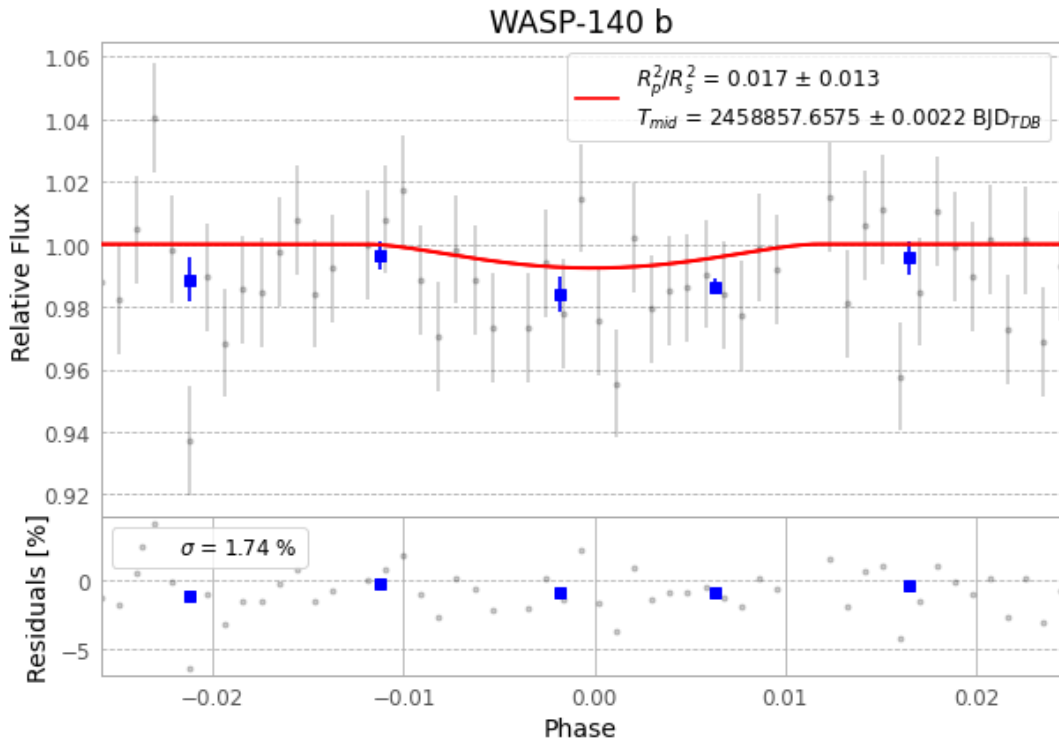


Figure 1 — FinalLightCurve_WASP-140 b_2020-01-08.png (SHA-256 541cd6e523b487b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [197, 322], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6af9aa7b3bd4dbb6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

56 FITS frames (37 MB), WASP-140200109021512.FITS → WASP-140200109050012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-200109.log (7845fb4f2ed0...), FinalLightCurve_WASP-140 b_2020-01-08.png (541cd6e523b4...), FinalLightCurve_WASP-140 b_2020-01-08.csv (11ded8e9dc55...)

Compute resources

Wall time 456.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:30Z.